

Multi-Class Classification of Crime Severity Levels Using Case Metadata and FIR Attributes

Abstract :

Crime is becoming more diverse and complex day by day. This challenges law enforcement agencies not only in investigation and prosecution but also in prioritizing cases according to their severity.

Classifying the severity has become a guessing game which is further influenced by factors like human bias, uneven practices, and overcrowded police stations.

This situation has a significant impact on ordinary people. When serious crimes get dismissed as minor crimes, families feel helpless, justice is delayed, and entire communities lose faith in our judicial as well as our police system.

This research examines crime severity levels across multiple Indian cities from 2020 to 2024, using data from a Kaggle available dataset. It includes detailed information about the FIR attributes and covers various offenses, including assault, kidnapping, identity theft, cybercrimes, homicide and many more.

An in depth analysis has been conducted on multiple machine learning models (XGBoost, Catboost, LightGBM, Decision Tree), ensemble stacking, and a deep learning MLP with resampling techniques. Across all the models, Direct XGBoost and OVR Catboost delivered best results, with their accuracy being 0.80 and Macro F1 Score being 0.72. To address the imbalance in the dataset we applied advanced resampling techniques like SMOTETomek and class weights.

Through this analysis, we found out which are the most crime prone regions in India. It also reports Delhi as a state with the highest number of crimes reported. When we divided India into regions, the west reported the highest number of crimes while the central reported the minimum number of crimes. Gender based analysis showed that women have a higher victimization rate.

Our findings suggest that FIR-based machine learning can not only predict how severe a crime might be but also guide law enforcement to prioritise severe crimes.

Keywords

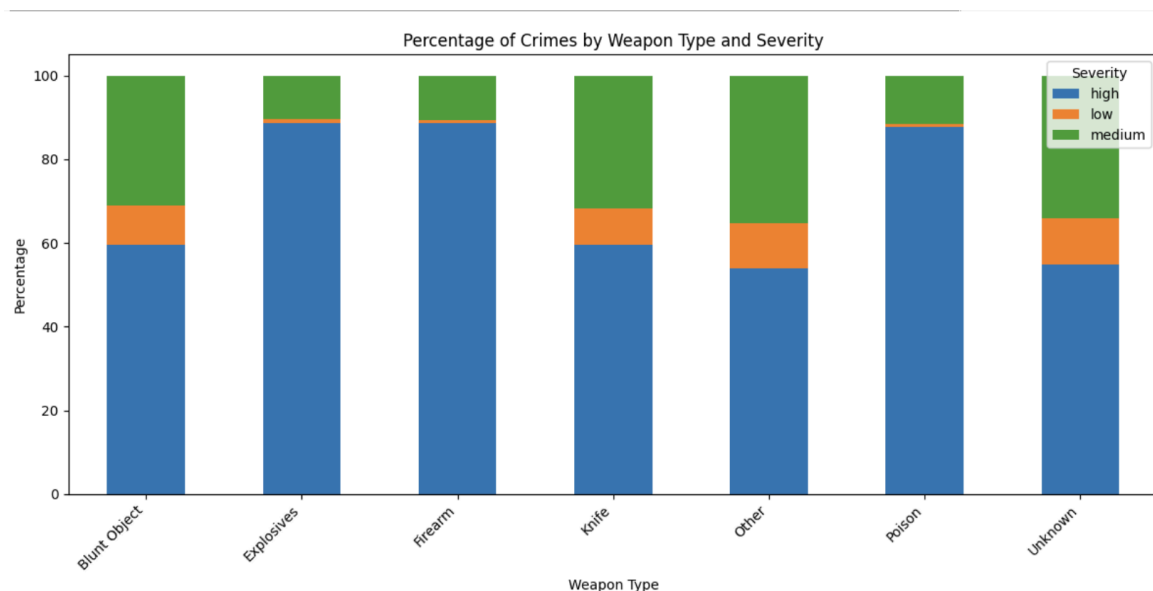
Crime severity classification, FIR attributes, case metadata, machine learning, imbalanced data, multi-class classification

Introduction :

Crime severity classification plays an important role in helping law enforcement and judicial processes. Despite many legal reforms, crime reporting and its categorization in India remains complex and influenced by cultural and social factors. Crime severity is essential to prioritize cases, most importantly in our overburdened justice system.

Fast Track Special Courts (FTSCs) were set up in India to speed up justice in sensitive cases. They were set up after the 2012 Nirbhaya Case, when there was huge public outrage for faster justice [24, 25]. They are unlike regular courts that often take years. These courts are meant to resolve severe cases within months, so victims don't have to wait so long for the closure.

In 2024, they resolved more than 85,000 sexual offence cases with a 96% disposal rate [24, 26]. Though these courts have resolved many cases, many serious crimes like murder and violent attacks still take years to reach a decision. The consequences of this issue are more wide. When justice is delayed to a very severe crime in our courts, victims wait years for closure and the neighbourhood feels unsafe [27]. This is why this study is significant.



As shown in Figure 1, our analysis reveals that high severity covers the maximum part across various weapons but low and medium severity have variations. Most previous works used general statistics like day, time, gender and crime type but FIR attributes and case metadata tells us the true nature of crime. These FIR attributes and case metadata are number of police personnel deployed, weapon used and crime description.

The weakness in the justice system is reflected by its low conviction rate as hearings are dragged on for years [16]. Researchers all around the world have been working to detect crime severity, which is why data analytics and machine learning models have become increasingly important in this effort. The gap in verdict of low severe and highly severe crimes reflects a big failure on case prioritization. This gap can be addressed by new AI, data mining and machine learning approaches. In this research, we have worked on machine learning and deep learning

approaches to predict crime severity level along with the help of case metadata and FIR attributes.

Research Objectives :

- a. To classify and rank crimes based on their severity by using FIR attributes and case metadata as parameters.
- b. To apply and evaluate ML algorithms like Catboost, XGBoost, LightBGM, Decision Trees, ensemble stacking and Deep Learning algorithms to predict level of severity.
- c. To help law enforcement agencies prioritize high severity crime so that justice is not delayed and guilty is punished as soon as possible, thus ensuring public safety.

Literature Review

Crime severity and its prediction have been studied in many different ways, using both machine learning and deep learning. Most of this work has focused on understanding crime patterns or predicting whether a crime will happen, but only a few actually go deep into *how severe* a crime is. Also, many previous works fail to use the right features for crime severity prediction. This section looks at what's been done earlier, the models researchers used, and the kinds of features that mattered most.

Some of the earliest work used city crime datasets. For example, in Boston crime data, Decision Trees gave around 73% accuracy, doing better than models like SVM, Logistic Regression, and Naive Bayes. The features here were time based: day, season, time, and crime type. This study suggested that season and time influence crime occurrence and its severity [1]. Other papers showed that Decision Trees or C4.5 also perform decently when demographic or offender details are available [6]. In some cases, people combined Naive Bayes, Apriori, and Decision Trees using inputs like whether a weapon or vehicle was involved [10]. On much larger city datasets like Chicago, San Francisco, and Philadelphia, tree-style models again came out on top [12].

What's clear is that there isn't a single perfect model. Accuracy keeps shifting depending on the dataset, which means the choice of data is just as important as the algorithm [1,6,7,10,12]. Some studies also added spatial features like GPS coordinates or time, which made KNN and Naive Bayes useful for hotspot mapping [2]. Random Forests and Decision Trees have been used in risk-scoring tools, but those usually predict the *likelihood* of crime, not how severe it will be [14].

There's also work using deep learning. LSTM models, for instance, help capture time patterns and can predict trends [9]. ANN or LSTM generally do better when datasets are really big [4,5,8,15]. Some studies even mix supervised and unsupervised methods — like clustering to group crimes before applying models [3]. Others use regression-style approaches (Random Forest Regression, Holt-Winters smoothing) to forecast crime over time [12,13].

Overall, tree-based models like Random Forest or XGBoost are consistently strong for tabular datasets, while deep learning has advantages for sequences or text-heavy fields [1,6,8,9,11,12,15]. But the gap is that not many studies have tried multi-class severity prediction. Rarely any study has used FIR attributes and case metadata as useful parameters.

Very few handle class imbalance directly. This shows that while crime prediction is a well-studied field, crime severity itself is still underexplored.

Methodology / Proposed Work

Data Collection and Preprocessing

This study has used an Indian crime dataset that we picked up from Kaggle [17]. This dataset captures a comprehensive snapshot of criminal activities across multiple Indian cities from 2020 to 2024. It includes information about the type of crime, the date and time of occurrence, the weapon used, victim demographics, and the level of police deployment [17]. With crimes ranging from identity theft to homicide, this dataset offers valuable insights for researchers, policymakers, and law enforcement agencies aiming to understand crime patterns and improve public safety. Figure 2 shows 5 columns of dataset :

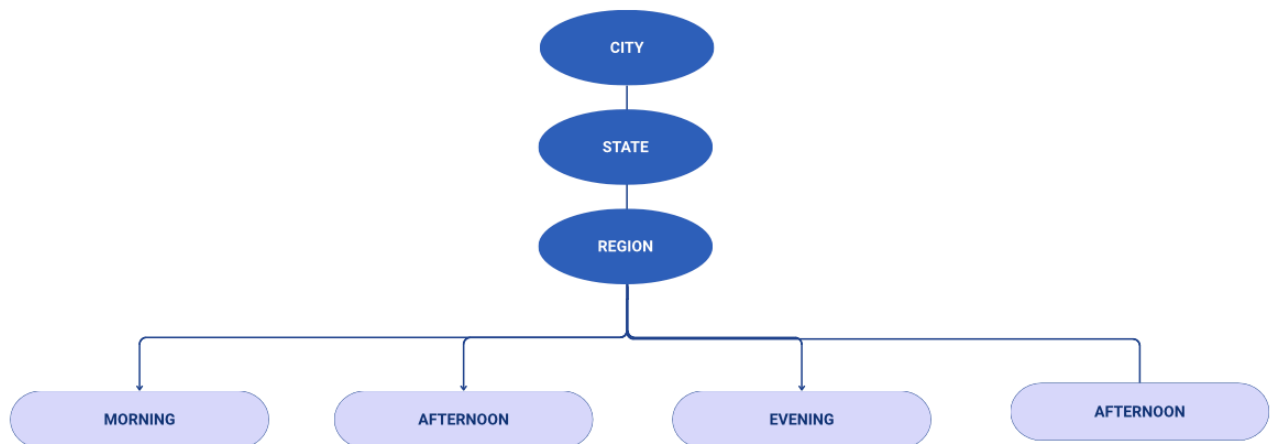
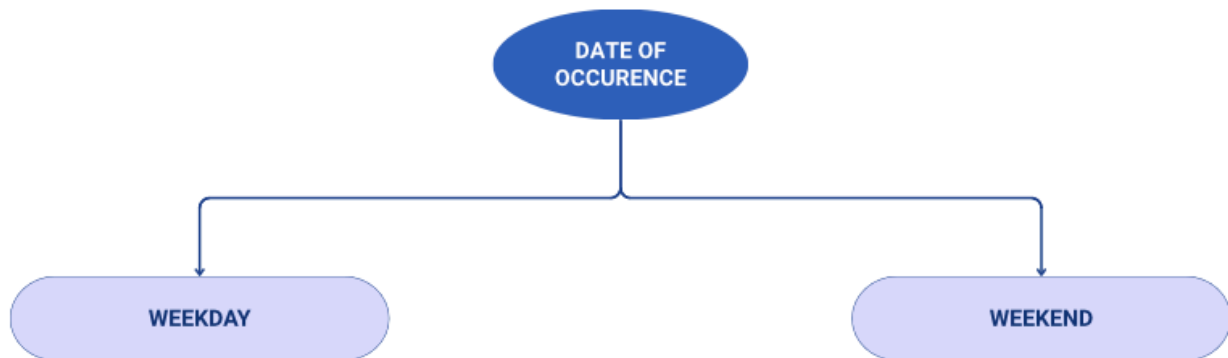
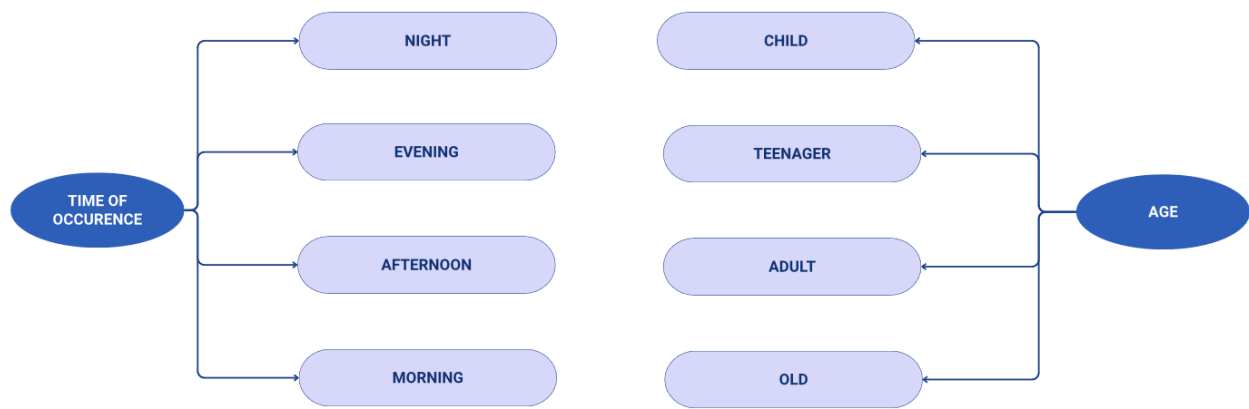
	Report Number	Date Reported	Date of Occurrence	Time of Occurrence	City	Crime Code	Crime Description	Victim Age	Victim Gender	Weapon Used	Crime Domain	Police Deployed	Case Closed	Date Case Closed
0	1	02-01-2020 00:00	01-01-2020 00:00	01-01-2020 01:11	Ahmedabad	576	IDENTITY THEFT	16	M	Blunt Object	Violent Crime	13	No	NaN
1	2	01-01-2020 19:00	01-01-2020 01:00	01-01-2020 06:26	Chennai	128	HOMICIDE	37	M	Poison	Other Crime	9	No	NaN
2	3	02-01-2020 05:00	01-01-2020 02:00	01-01-2020 14:30	Ludhiana	271	KIDNAPPING	48	F	Blunt Object	Other Crime	15	No	NaN
3	4	01-01-2020 05:00	01-01-2020 03:00	01-01-2020 14:46	Pune	170	BURGLARY	49	F	Firearm	Other Crime	1	Yes	29-04-2020 05:00
4	5	01-01-2020 21:00	01-01-2020 04:00	01-01-2020 16:51	Pune	421	VANDALISM	30	F	Other	Other Crime	18	Yes	08-01-2020 21:00

To derive our desired information, we have modified the data in following ways:

- Removed irrelevant columns such as 'Report Number', 'Date Reported', 'Crime Code', 'Case Closed', and 'Date Case Closed' and a total of 9 attributes were selected, which are shown here. Figure 3 shows table with selected attributes :

	Date of Occurrence	Time of Occurrence	City	Crime Description	Victim Age	Victim Gender	Weapon Used	Crime Domain	Police Deployed
0	01-01-2020 00:00	01-01-2020 01:11	Ahmedabad	IDENTITY THEFT	16	M	Blunt Object	Violent Crime	13
1	01-01-2020 01:00	01-01-2020 06:26	Chennai	HOMICIDE	37	M	Poison	Other Crime	9
2	01-01-2020 02:00	01-01-2020 14:30	Ludhiana	KIDNAPPING	48	F	Blunt Object	Other Crime	15
3	01-01-2020 03:00	01-01-2020 14:46	Pune	BURGLARY	49	F	Firearm	Other Crime	1
4	01-01-2020 04:00	01-01-2020 16:51	Pune	VANDALISM	30	F	Other	Other Crime	18

- The presence of missing values was addressed by entering the 'unknown' in the instances of Weapon Used and Victim Gender as well as replacing the missed values of the Victim Age with the median age.
- Date and Time attributes were used and classified by Day (Weekend/Weekday) and Time of Day (Night, Morning, Afternoon, Evening, Late Night)
- States were also mapped on to the cities, which in turn were grouped into greater regions (North, South, East, West, Central) due to the area based data.
- The ages of the victims were grouped into following originages : Child, Teenager, Adult, and Senior.



- As the dataset did not contain a severity label, we created our custom severity column as further described in '**Constituting the model section**'.
- The features of interaction, like Crime_Region, Age_TimeOfDay, and Weapon_TimeOfDay, were introduced to enhance the input of the model and the categorical type features were determined to be encoded afterward.
- All categorical variables were encoded with One-hot to train models.
- Addressing data imbalance :
 1. Applied Smote but failed to Mediate on the imbalance of classes

2. SMOTETomek was used and weight was also produced on classes leading to a balanced dataset.

Before data balancing :

Class	
0	6779
1	582
2	2277

After Data balancing with SMOTETomek :

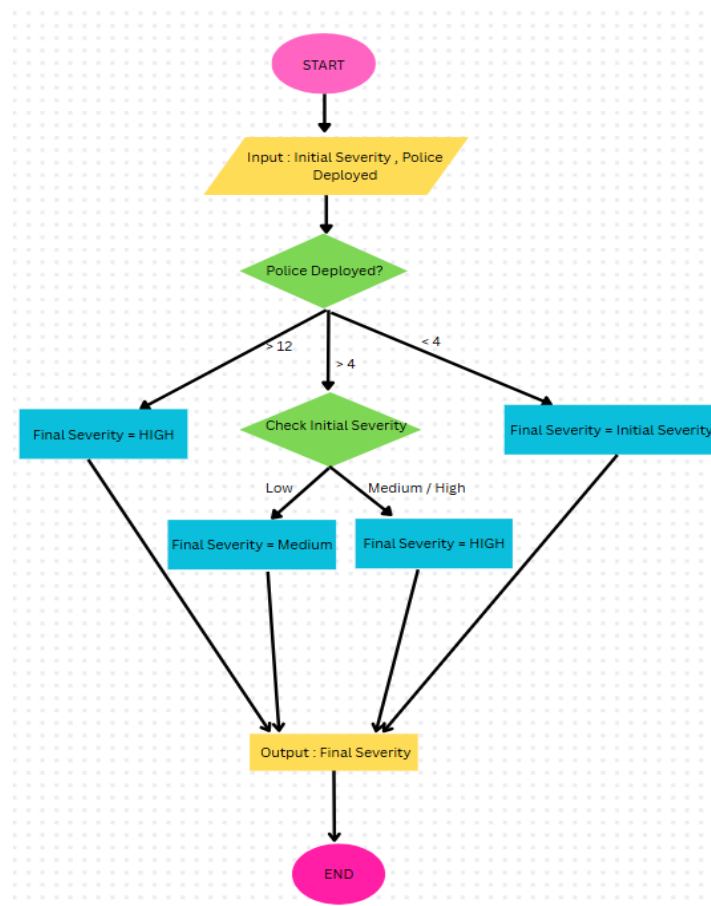
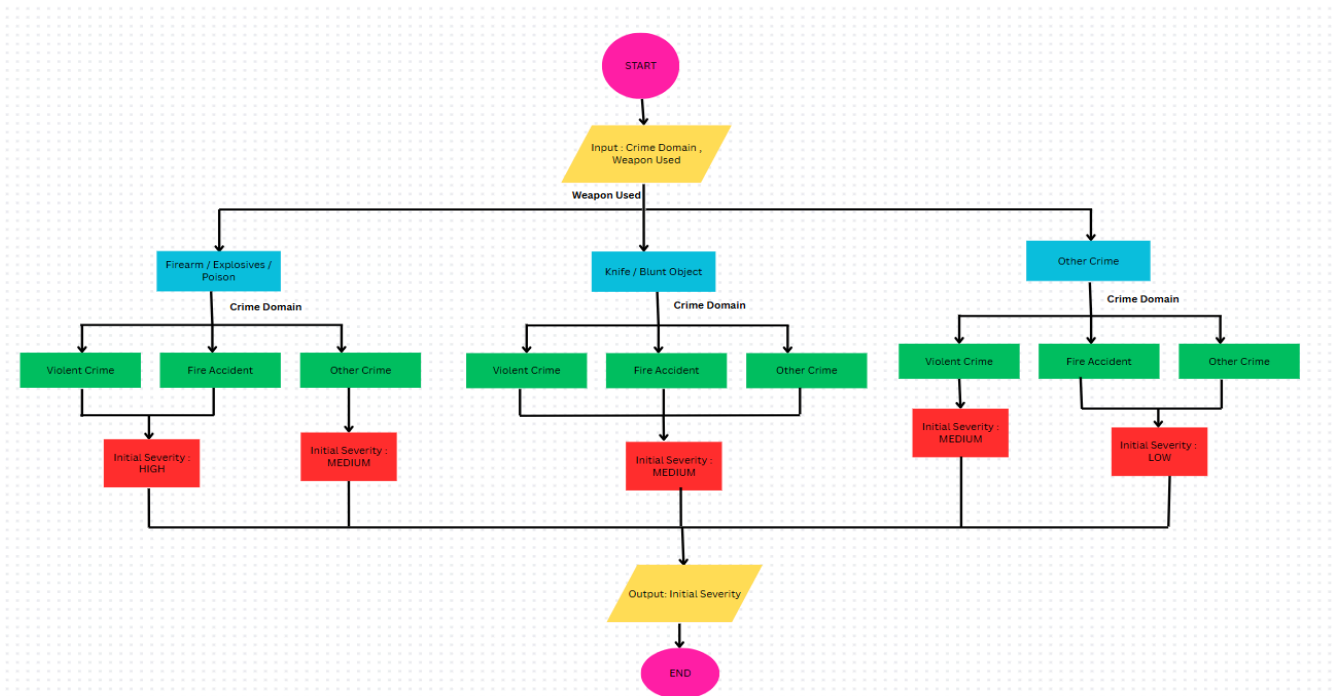
Class	
0	6601
1	6747
2	6585

After Data balancing with Class Weights :

Class	
0	1.0066
1	0.9848
2	1.0090

Constituting The Model (Severity Column Preparation) :

The dataset did not have any direct severity labels, so we made a custom multi-class severity classification to categorize crimes into **low, medium, and high severity**. The severity was determined based on three key factors: **crime domain, weapon used**, and the **number of police personnel deployed**.



This two - part flowchart defines the crime's severity. The first part uses the weapon and crime type to establish an initial severity label (low , medium , high) . The second part refines this by considering police deployment , which can escalate the initial severity to a final severity level.

Crime Domain:

- *Violent crimes* were treated as higher severity compared to “Other Crime” or “Traffic Fatalities.”
- *Fire accidents* and *traffic fatalities* were separately judged due to their distinct contexts.

Weapon Used:

- Weapons such as **firearms, explosives, or poison** were categorized as *high severity*.
- Weapons such as **knives or blunt objects** were categorized as *medium severity*.
- Cases involving **unknown, other, or undefined weapons** were categorized as *low severity*.

Police Deployment:

- **The number of police deployed was considered an important indicator because higher deployment reflects greater crime severity.**
- Deployment of **≥12 officers** automatically turned the incident to *high severity*.
- Deployment of **4–11 officers** increased severity by one level (e.g., low → medium, medium → high).

Crime Prediction :

In this paper, we are using four cutting edge machine learning models : Catboost, XGBoost, LightGBM and Decision Trees, along with a stacking ensemble approach. We also used a deep learning model - keras baseline. Each model has its own advantages.

- CatBoost: Chosen for its strength in handling categorical variables directly and reducing the need for heavy preprocessing.
- XGBoost: Chosen due to its high accuracy and proven performance on imbalanced datasets.
- LightGBM: Offers efficiency and scalability in big datasets.
- Decision Tree: Used as a simple baseline to compare with more complex models.
- Stacking Ensemble: Integrated to leverage complementary strengths of multiple models for better generalization.
- Keras Baseline: Helps in capturing deeper feature interactions.

- **CatBoost (OVR Tuned)**

Catboost is a gradient boosting model based on decision trees. It is developed to handle categorical values while reducing overfitting and prediction bias, the problem that we faced a lot in our dataset. It uses ordered boosting to maintain unbiasedness in result. This makes it suitable for tabular datasets with both numerical and categorical columns [18].

With catboost we applied One-vsRest (OVR) strategy to tackle multi class crime severity prediction. Hyperparameter tuning was performed using Optuna to maximize Macro F1 Score.

- **XGBoost (Tuned)**

XGBoost (Extreme Gradient Boosting) is an efficient and scalable implementation of gradient boosting that has gained popularity for structured/tabular data tasks. It uses boosting with regularization to reduce overfitting and can run processes in parallel, which helps it train faster and work well for tasks like classification and regression[19]. For this study, we implemented a direct multi-class XGBoost model to predict crime severity. Hyperparameter tuning was carried out using Optuna to maximize the Macro F1 score.

- **LightGBM (OVR Default Base)**

LightGBM (Light Gradient Boosting Machine) is a gradient boosting framework that uses tree-based learning algorithms. It is designed in such a way that it is highly efficient with large datasets [20]. It uses gradient boosting techniques in order to build a decision tree. It is known for its scalability and speed, making it a right choice for processing of our dataset. [mam]

In this study, we applied One-vsRest (OVR) strategy to tackle multi class crime severity prediction. Hyperparameter tuning was carried out using Optuna to maximize the Macro F1 score.

- **Stacking**

In this research, we tried ensemble learning methods like stacking as well. In stacking, multiple models are combined to give better predictions. It takes output from several base models and feeds them into a higher-level-meta-modela which learns how to make final predictions. This technique can balance out weakness of individual models and also performs well on tabular data. [21]

For this work, we built two stacking classifiers. The base models consisted of CatBoost, XGBoost, and Random Forest. In the first classifier, a Random Forest worked as the meta-model *to capture complex non-linear patterns*. In the second, we used Logistic Regression as the final estimator for its simplicity and interpretability. We used Optuna to tune the hyperparameters as it helps in optimization and stability.

Although stacking showed benefits but its performance was slightly lower than XGBoost and CatBoost individually.

- **Decision Tree (Tuned)**

True to name, the decision trees algorithm forms one of the popular methods utilized to accomplish a specific objective by using a tree structure of decisions [1]. It is a well-known and utilized machine learning approach to both classification and regression. The model operates by repeatedly partitioning the data based on features value, and breaking them down into the leaves of a tree in which each leaf corresponds to a predicted category. Since Decision Trees can be interpreted easily and can deal with both numerical and categorical features, they find application in crime prediction research [22].

To apply this algorithm, the three aspects were considered: the choice of the significant attributes according to which the tree will grow, the choice of the attribute on which the

tree will not be divided further, and the correct number of instances to ensure that the model does not become overfitted. [1]

In this paper, we used a Decision Tree to learn the encoded and resampled data. To determine the optimal values of maximum depth, minimum samples in a leaf, and splitting criterion, hyperparameter tuning was used on a grid-search CV and cross-validation. This was done in order to optimize the model and prevent over fitting.

- **Keras Baseline MLP**

The Keras Baseline model is a Multi-Layer Perceptron (MLP), a type of neural network used for classification tasks. [23] MLPs consist of multiple fully connected layers with activation functions that allow the model to learn complex, non-linear relationships in the data. They work very well for tabular datasets that have both categorical and numerical features [23]. So it becomes suitable for this problem statement.

In this study, the baseline MLP was trained using the resampled dataset obtained from SMOTETomek to address class imbalance. Class weights were applied during training to further balance the impact of minority and majority classes. The architecture included :

- Several dense layers with ReLU activation
- A softmax output layer for multi-class classification

Then the model was trained using categorical cross-entropy loss along with the Adam optimizer. Hybrid sampling and focal loss techniques were also applied to increase performance on minority classes.

Results and Discussion

Table of results

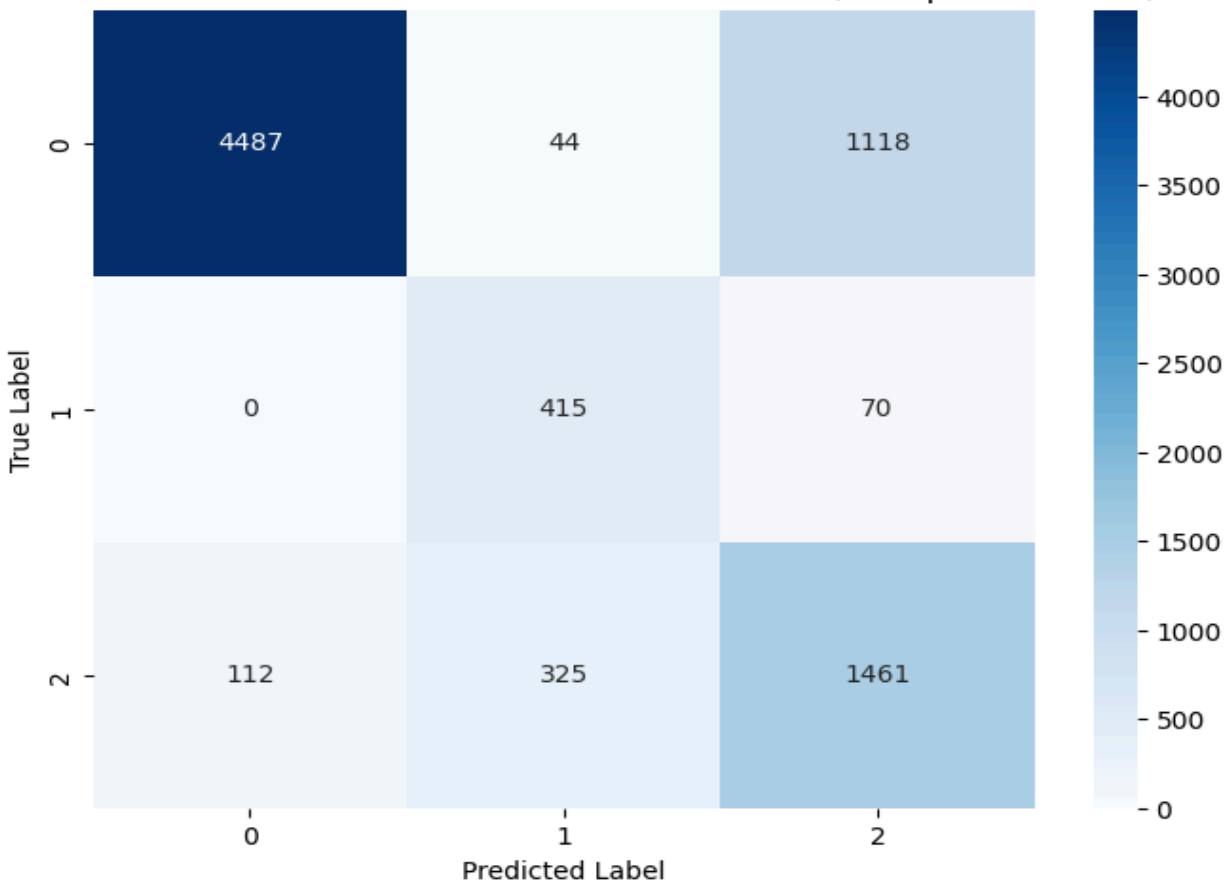
Model	Accuracy	Macro F1	F1 (Class 0)	F1 (Class 1)	F1(Class 2)
CatBoost (OvR Tuned)	0.79	0.72	0.87	0.64	0.65
XGBoost (Tuned)	0.80	0.72	0.87	0.65	0.64
LightGBM (OvR Default Base)	0.79	0.72	0.88	0.64	0.64
Stacking Classifier LR Final (Tuned)	0.79	0.70	0.88	0.64	0.62

Decision Tree (Tuned)	0.78	0.69	0.87	0.61	0.60
Keras Baseline	0.79	0.71	0.88	0.65	0.60

Analysis of result

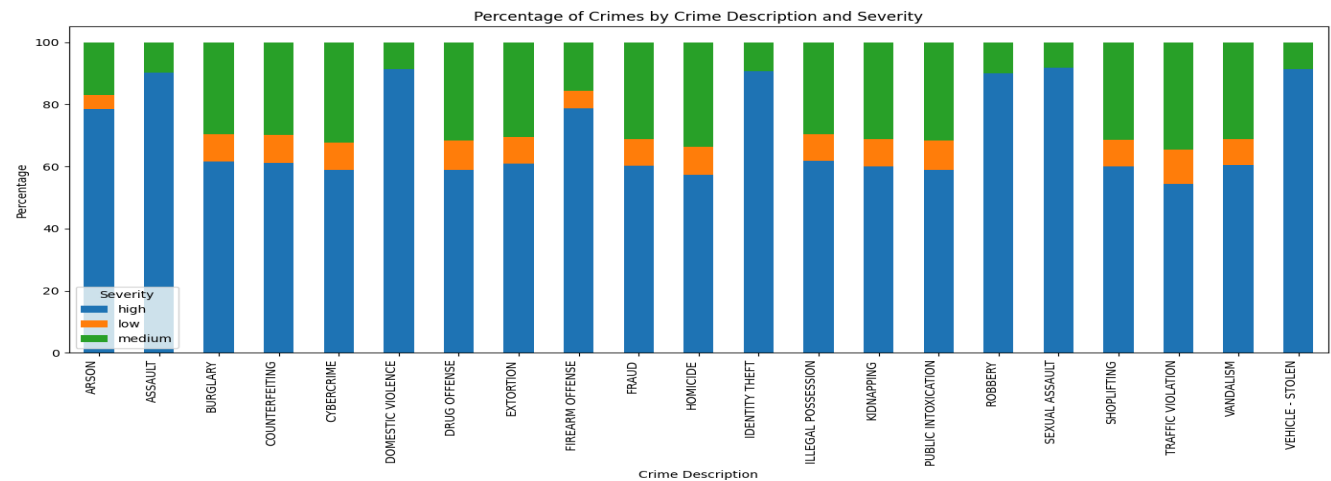
In this section, all the results of the implementations of the models we discussed in the previous section are presented. Model performance reflects these underlying patterns in the dataset. The Direct XGBoost (Tuned) model had the highest accuracy of 0.80 and Macro F1 of 0.72, followed by OVR CatBoost (Tuned) which had 0.79 accuracy and 0.72 Macro F1. OVR LightGBM (Default Base) had 0.79 accuracy and 0.72 Macro F1. The stacking classifier (LR Final, Tuned) had 0.79 accuracy and 0.70 Macro F1, demonstrating an advantage of ensembling but worse than single gradient boosting models. The Decision Tree (Tuned) was trained with 0.78 accuracy and 0.69 Macro F1 which is a simpler baseline. The Keras Baseline MLP with focal loss recorded a 0.79 accuracy and 0.71 Macro F1, which shows better performance in all classes and recall on minority severity classes. However, XGBoost required less time to provide results and also gave the best results compared to all other models.

Confusion Matrix for Direct Multi-class XGBoost Model (from provided data)

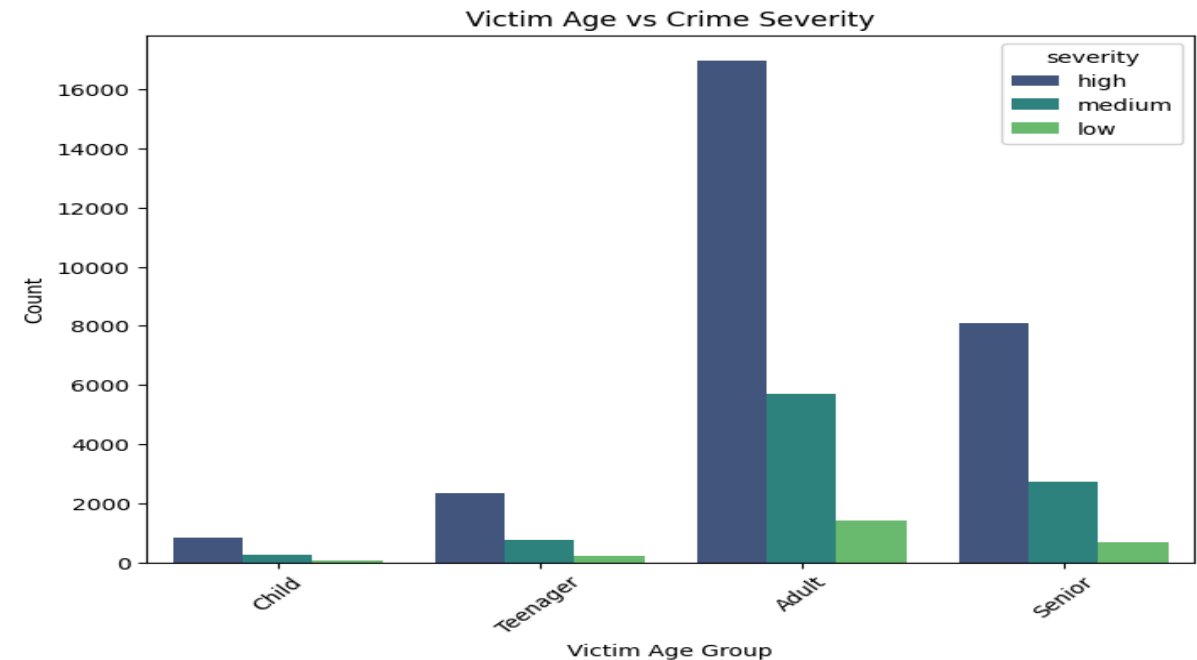


The Macro F1 scores, our main focus due to the unbalanced dataset, show that gradient boosting models (XGBoost, CatBoost, LightGBM) performed best with 0.72, while simpler models like Decision Tree lagged at 0.69. Most models did well on Class 0, but performance dropped for Classes 1 and 2, highlighting the difficulty of predicting higher severity crimes.

The results support the idea that FIR attributes, case metadata, and machine learning algorithms are important to successfully predicting crime severity. This crime dataset analysis identifies some key characteristics that give an insight into the factors that have a high impact on crime severity. An example of such a feature in our dataset is ‘Crime Description’. The description of the crime was identified as the most correlated with the severity, so it can be concluded that the nature of the crime is the strongest predictor of its impact.

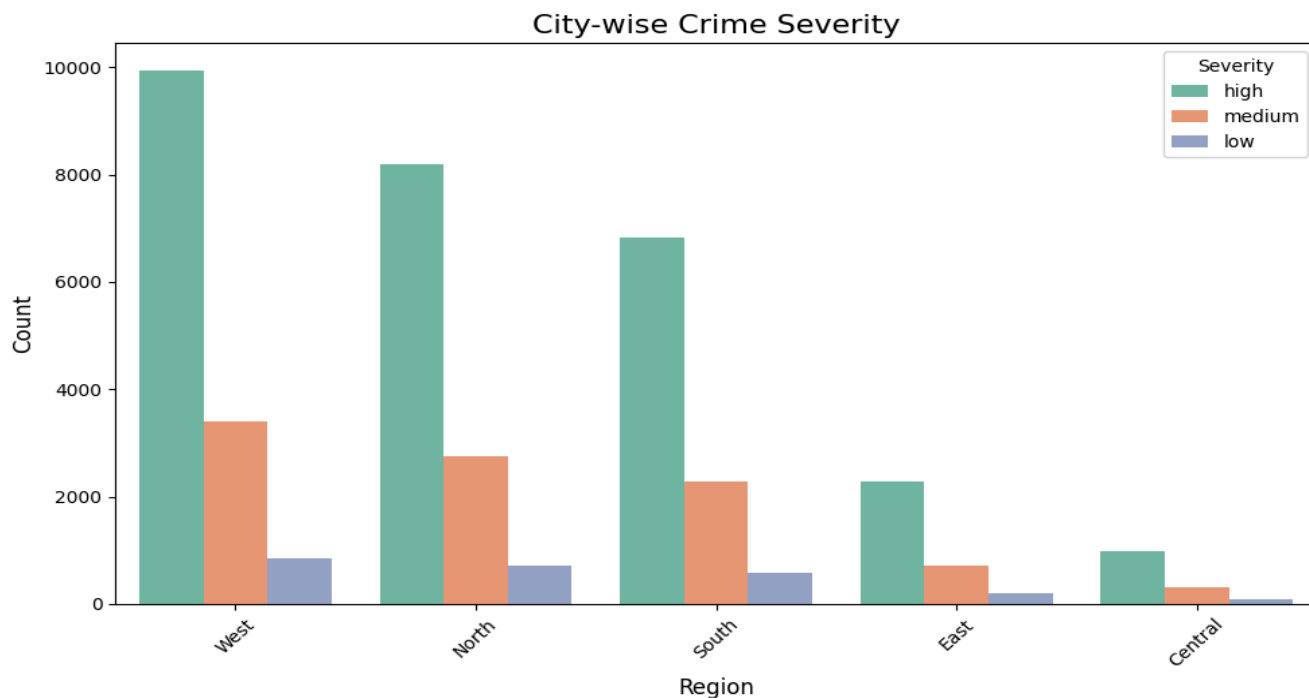


The high-severity crimes prevail in the dataset, and **adults**, followed by seniors are more likely to be targeted compared to children and teenagers.



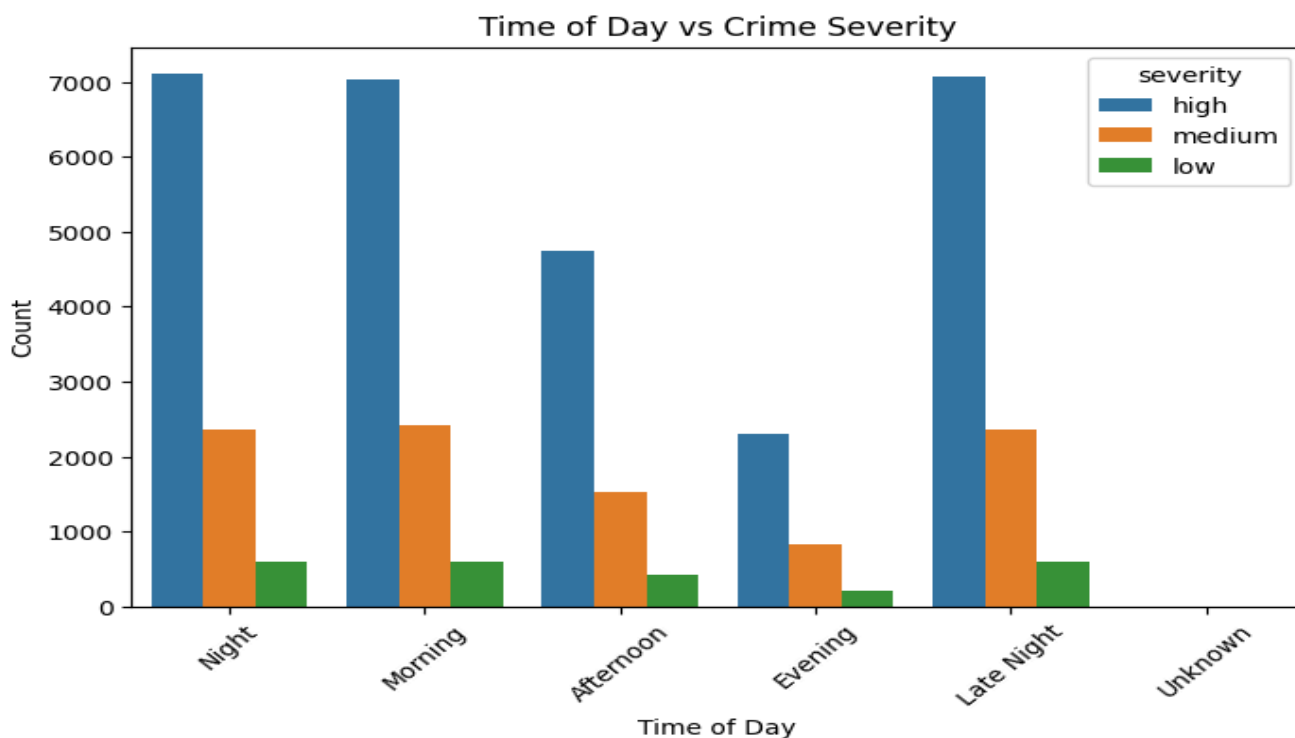
The West region has the highest numbers of crimes of all the levels of severity, followed by the North and the South, whereas the Central region has the lowest crime levels.

(visualization of region)



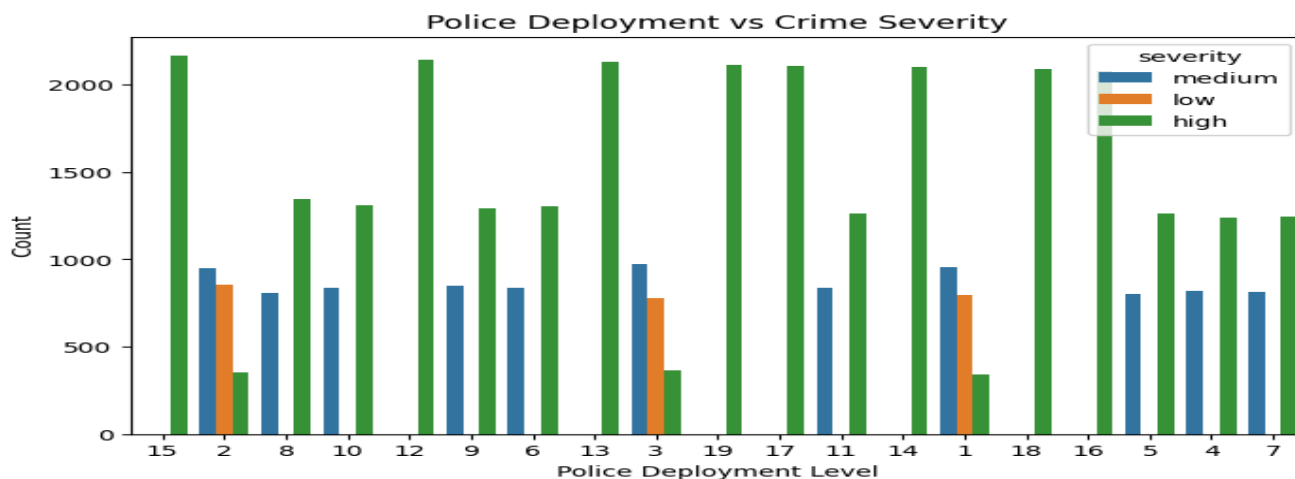
Temporal patterns show that the probability of severe crimes is higher during the night and it is not surprising, but high-severity crimes are constantly high and there is a slight increase in May.

(visualisation of time)



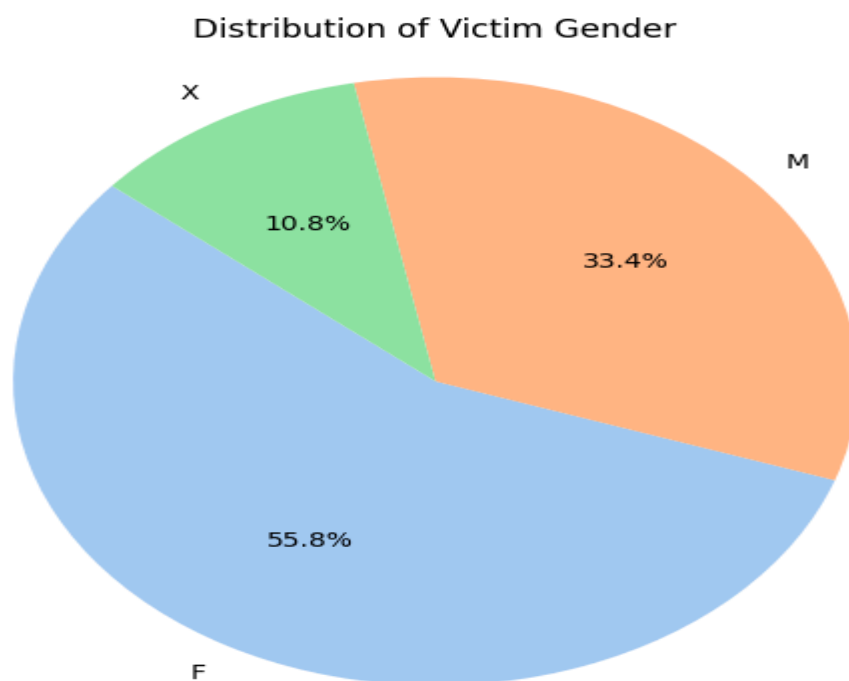
Also, police deployment is also an important factor in severity as mentioned earlier since more officers are needed in the incidents that are considered as high-severity.

(visualization of distribution of crime severity acc to pd)



The gender distribution of the victims in our research indicates that a large part of the data belongs to women, which means that men and women are vulnerable in different types of crimes.

(visualisation of gender)



These results are not limited to model performance, they demonstrate how FIR attributes can be used directly to prioritize cases. This can be used to decrease time of delay in severe cases and help in quick justice by correlating these severity predictions to actual police records.

Visualization of result

Confusion matrix of every model if space available.

Conclusion and Future Work

Our study shows that looking closely at FIR details along with case records can play a big role in predicting how severe a crime is when using machine learning and deep learning. These case records connect predictions with the records police rely on every day, which makes our result more practical and relevant to real life crime situations. This information was often not considered in previous research works.

We created a custom severity label as the target column was not present in the dataset and tested both machine learning and deep learning models. In general, machine learning models worked well. As shown in results, XGBoost and Catboost with OvR performed consistently well. They both reached accuracy close to 0.80 and Macro F1 scores above 0.72. In these models, we found that using XGBoost was practical for real-time deployment because it trained the fastest. We also noticed that the strongest indicators of severity level were crime description, number of police personnel deployed and weapon used.

Future studies can extend this work by improving the performance of the models further. They can work on increasing the Macro F1 score and reducing biases across classes. A fairer balance can be achieved in high, medium and low severity predictions if we further work on advanced techniques, hybrid models and more advanced resampling techniques.

The other area in which we will be working is the use of a more balanced dataset. The data used in the research was also biased in its distribution of levels of severity of crimes that restricted the generalization of the model. The use of fresher and more detailed FIR data can provide a good foundation on which to build models that would be more reliably applicable across crime types.

This research takes a step toward data-driven justice, showing how FIR records can be transformed into practical tools for faster, fairer case prioritization.

Acknowledgement

We thank Dr. Ritu and Dr. Garima Jaiswal for their guidance and Indira Gandhi Delhi technical University for their support.

References

- [1] S. Tasnim, P. Sarkar, A. Hossain, and M. A. Ali, "A classification approach to predict severity of crime on Boston city crime data," in *Proc. 7th Int. Conf. Data Science & SDGs*, Rajshahi, Bangladesh, Dec. 2019, pp. 405–412.
- [2] H. K. ToppiReddy, B. Saini, and G. Mahajan, "Crime prediction & monitoring framework based on spatial analysis," *Procedia Comput. Sci.*, vol. 132, pp. 696–705, 2018, doi: 10.1016/j.procs.2018.05.075.

- [3] S. R. Bandekar and C. Vijayalakshmi, "Design and analysis of machine learning algorithms for the reduction of crime rates in India," *Procedia Comput. Sci.*, vol. 172, pp. 122–127, 2020, doi: 10.1016/j.procs.2020.05.018.
- [4] R. M. Saeed and H. A. Abdulmohsin, "A study on predicting crime rates through machine learning and data mining using text," *J. Intell. Syst.*, vol. 32, no. 1, 2023, doi: 10.1515/jisys-2022-0223.
- [5] E. C. Ateş, "Big data, data mining, machine learning, and deep learning concepts in crime data," *J. Penal Law Criminol.*, pp. 293–319, 2021, doi: 10.26650/jplc2020-813328.
- [6] O. Llah, "Crime analysis and prediction using machine learning," in *Proc. 43rd Int. Conv. Inf., Commun. Electron. Technol. (MIPRO)*, 2020, pp. 496–501, doi: 10.23919/MIPRO48935.2020.9245120.
- [7] L. McClendon and N. Meghanathan, "Using machine learning algorithms to analyze crime data," *Mach. Learn. Appl.: Int. J.*, vol. 2, no. 1, pp. 1–12, 2015, doi: 10.5121/mlaij.2015.2101.
- [8] N. H. Shamsuddin, N. A. Ali, and R. Alwee, "An overview on crime prediction methods," in *Proc. 6th ICT Int. Student Project Conf. (ICT-ISPC)*, 2017, pp. 1–5, doi: 10.1109/ICT-ISPC.2017.8075335.
- [9] M. Manengadan, S. Nandan, and N. Subash, "Crime data analysis, visualization and prediction using LSTM," *Int. J. Data Sci. Anal.*, vol. 7, no. 3, pp. 51–58, 2021, doi: 10.11648/j.ijdsa.20210703.11.
- [10] S. Sathyadevan, M. S. Devan, and S. Gangadharan, "Crime analysis and prediction using data mining," Amrita Center for Cyber Security, Amrita Vishwa Vidyapeetham, Kerala, India, 2014.
- [11] W. Safat, S. Asghar, and S. A. Gillani, "Empirical analysis for crime prediction and forecasting using machine learning and deep learning techniques," *IEEE Access*, vol. 9, pp. 70080–70094, 2021, doi: 10.1109/ACCESS.2021.3078117.
- [12] M. Feng, et al., "Big data analytics and mining for crime data analysis, visualization and prediction," in *Lecture Notes in Computer Science*, pp. 605–614, 2018, doi: 10.1007/978-3-030-00563-4_59.
- [13] R. M. Aziz, et al., "Machine learning-based soft computing regression analysis approach for crime data prediction," *Karbala Int. J. Mod. Sci.*, vol. 8, no. 1, pp. 1–19, 2022, doi: 10.33640/2405-609X.3197.
- [14] M.-S. Baek, et al., "Smart policing technique with crime type and risk score prediction based on machine learning for early awareness of risk situation," *IEEE Access*, vol. 9, pp. 131906–131915, 2021, doi: 10.1109/ACCESS.2021.3112682.
- [15] P. Saravanan, et al., "Survey on crime analysis and prediction using data mining and machine learning techniques," in *Lecture Notes in Electrical Engineering*, pp. 435–448, 2020, doi: 10.1007/978-981-15-7241-8_31.
- [16] Ritika, et al., "Predictive analysis of crime against women in India: Uncovering critical insights," *Women Crim. Justice*, pp. 1–38, 2025, doi: 10.1080/08974454.2025.2543087.

- [17] SudhanvaHG, "Indian crimes dataset," Kaggle, 2024. [Online]. Available: <https://www.kaggle.com/datasets/sudhanvahg/indian-crimes-dataset>. [Accessed: July 01, 2025].
- [18] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*, 2018, pp. 6638–6648. [Online]. Available: <https://papers.nips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>.
- [19] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, San Francisco, CA, USA, Aug. 2016, pp. 785–794, doi: 10.1145/2939672.2939785.
- [20] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017, pp. 3146–3154. [Online]. Available: <https://papers.nips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>.
- [21] D. H. Wolpert, "Stacked generalization," *Neural Netw.*, vol. 5, no. 2, pp. 241–259, 1992, doi: 10.1016/S0893-6080(05)80023-1.
- [22] J. R. Quinlan, *C4.5: Programs for Machine Learning*. San Mateo, CA, USA: Morgan Kaufmann, 1993.
- [23] F. Chollet, "Keras," 2015. [Online]. Available: <https://keras.io>. [Accessed: Aug. 19, 2025].
- [24] "Fast track special courts," Drishti IAS, 22 Mar. 2025. [Online]. Available: <https://www.drishtiias.com/daily-updates/daily-news-analysis/fast-track-special-courts-2>. [Accessed: Aug. 19, 2025].
- [25] "Swift justice, safer society: The impact of fast track special courts," Press Information Bureau, Ministry of Law and Justice, Government of India, 20 Mar. 2025. [Online]. Available: <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2113344>. [Accessed: Aug. 19, 2025].
- [26] "Swift justice, safer society: The impact of fast track special courts," Lukmaan IAS Blog, 3 Apr. 2025. [Online]. Available: <https://blog.lukmaanias.com/2025/04/03/swift-justice-safer-society-the-impact-of-fast-track-special-courts/>. [Accessed: Aug. 19, 2025].
- [27] G. Pandey, "The unsettling consequences of justice delay in India: A grave situation with alarming data," *The Legal Perspective*, Times of India (readers' blog), 11 Jun. 2023. [Online]. Available: <https://timesofindia.indiatimes.com/readersblog/the-legal-perspective/the-unsettling-consequences-of-justice-delay-in-india-a-grave-situation-with-alarming-data-55042/>. [Accessed: Aug. 19, 2025].